INJECTIVITY OF QUADRATIC DIVISOR TRANSFORMS ON ROOTS OF UNITY: PARAMETER CLASSIFICATION, ROTATION OBSTRUCTION, AND THE GALOIS BOUNDARY

Abstract. For a multiset S of roots of unity with power-sum function $p_S(m) = \sum_{\lambda \in S} \lambda^m$, we study the quadratic divisor transform $g^S_{a,b}(m) = p_S(am) \cdot p_S(bm)$ and the question of when it determines S.

We prove a complete parameter classification: $g_{a,b}$ is injective on the class of cycle-closed multisets (unions of complete sets of d-th roots of unity) if and only if $\min(a, b) = 1$. The positive direction is established by an explicit telescoping inversion formula exploiting p-adic stabilization on the divisor lattice. The negative direction is established by two families of counterexamples, one for each case of $\gcd(a, b)$.

We identify a rotation obstruction: if $\omega^{a+b} = 1$, then $g^{\omega S}_{a,b} = g^S_{a,b}$ for all S. Cycle-closed multisets are immune to this obstruction, and we prove they are precisely the Galois-stable multisets in μ_N—those fixed by $\mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$. The maximal domain of injectivity for $g_{1,k}$ is strictly larger than the cycle-closed class; its characterization is an open problem.

1. Introduction

1.1. **The transform.** Let $S = \{\lambda_1, \ldots, \lambda_n\}$ be a multiset of complex numbers. The *power-sum function* of S is $p_S(m) = \sum_{j=1}^n \lambda_j^m$. The *quadratic divisor transform* at speed pair (a, b) is

$$g^S_{a,b}(m) \;=\; p_S(am) \cdot p_S(bm).$$

Main question. For which pairs (a, b) and which classes of multisets S does $g^S_{a,b}$ determine S uniquely?

1.2. **Spectral origin.** For a permutation π with permutation matrix A, the synchronized tensor operator $S_{a,b} = A^{\otimes a} \otimes A^{\otimes b}$ satisfies $\mathrm{tr}(S^m_{a,b}) = g^S_{a,b}(m)$ where S is the eigenvalue multiset of A. Spectral determinacy of $S_{a,b}$ is thus equivalent to injectivity of $g_{a,b}$.

1.3. **Cycle-closed multisets.**

Definition 1.1. A multiset S of complex numbers is *cycle-closed* if

$$S = \bigsqcup_{i=1}^{r} \mu_{d_i},$$

where $\mu_d = \{e^{2\pi i k/d} : k = 0, \ldots, d-1\}$ is the complete set of d-th roots of unity.

Equivalently, S is the eigenvalue multiset of a permutation with cycle type $(d_1, \ldots, d_r)$.

1.4. **The divisor-sum function.** For a cycle-closed multiset with cycle type $(d_1, \ldots, d_r)$, writing $c_d = |\{i : d_i = d\}|$,

$$p_S(m) = \sum_{d|m} d \cdot c_d.$$

This is a non-negative, integer-valued, finitely-supported divisor-sum function.

1.5. **Summary of results.** The results decompose into two axes.

Axis 1 (Parameters): Complete classification. The transform $g_{a,b}$ is injective on cycle-closed multisets if and only if $\min(a,b) = 1$. Both directions are proved: Theorem 2.1 gives an explicit inversion formula when $\min(a,b) = 1$; Theorem 3.1 gives explicit counterexamples when $\min(a,b) \geq 2$.

Axis 2 (Domain): Sharp sufficient condition. The cycle-closed class is a natural and sufficient domain for injectivity, but it is not the maximal one. The rotation obstruction (Theorem 5.1) and Galois structure (Theorem 6.2) explain all computationally observed collisions among non-cycle-closed multisets, but do not generate collisions for every non-cycle-closed multiset. The maximal domain of injectivity remains open.

2. Telescoping Inversion

Theorem 2.1 (Telescoping Inversion)**.** *For every $k \geq 1$, the transform $g_{1,k}(m) = f(m) \cdot f(km)$ is injective on non-negative finitely supported divisor-sum functions. The inversion formula is:*

$$f(m) = \begin{cases} 0 & \text{if } g(m) = 0, \\ \sqrt{g(k^{2J}m)} \cdot \prod_{j=0}^{J-1} \dfrac{g(k^{2j}m)}{g(k^{2j+1}m)} & \text{if } g(m) > 0, \end{cases}$$

where $J = \min\{j \geq 0 : g(k^{2j}m) = g(k^{2j+1}m)\}$.

Lemma 2.2 (Monotonicity)**.** *$f(m) \leq f(km)$ for all $m, k \geq 1$.*

Proof. Every divisor of m divides km, and all coefficients c_d are non-negative. □

Corollary 2.3. *$g(m) = 0$ if and only if $f(m) = 0$.*

Lemma 2.4 (Stabilization)**.** *For each m, the sequence $f(k^j m)$ eventually stabilizes at a value $F_k(m)$ satisfying $F_k(km) = F_k(m)$.*

Proof. A cycle length d contributes to $f(k^j m)$ if and only if $d \mid k^j m$. For primes $p \mid k$: the condition $p^{v_p(d)} \mid k^j m$ is eventually satisfied as j grows. For primes $p \nmid k$: the condition $p^{v_p(d)} \mid m$ is independent of j. Hence the set of contributing cycle lengths stabilizes. The identity $F_k(km) = F_k(m)$ holds because multiplying by k only affects primes already saturated. □

Lemma 2.5 (Ratio Telescoping)**.** *When $f(m) > 0$:*

$$\frac{g(m)}{g(km)} = \frac{f(m) \cdot f(km)}{f(km) \cdot f(k^2 m)} = \frac{f(m)}{f(k^2 m)}.$$

The product telescopes:

$$\prod_{j=0}^{J-1} \frac{g(k^{2j}m)}{g(k^{2j+1}m)} = \frac{f(m)}{f(k^{2J}m)} = \frac{f(m)}{F_k(m)}.$$

Proof of Theorem 2.1. Combining the lemmas: $f(m) = F_k(m) \cdot \prod_{j=0}^{J-1} \frac{g(k^{2j}m)}{g(k^{2j+1}m)}$. The stabilized value satisfies $g(k^{2J}m) = F_k(m)^2$, so $F_k(m) = \sqrt{g(k^{2J}m)}$, and J is detectable as the first index where $g(k^{2j}m) = g(k^{2j+1}m)$. All quantities are computable from g alone. □

3. Failure for $\min(a,b) \geq 2$min(a,b) >= 2

Theorem 3.1. *For every (a,b) with $\min(a,b) \geq 2$, the transform $g_{a,b}$ is not injective on cycle-closed multisets.*

Proof. We treat two cases.

Case 1: $\gcd(a,b)=1$. Set $n=2ab$. Consider two cycle types: $\lambda=(ab,1^{ab})$ and $\mu=(a^b,b^a)$. Both have size $2ab$. They are distinct since λ has ab fixed points while μ has none.

For the collision, we compute the divisor-sum function at multiples of a:

$$f_\lambda(am)=1\cdot ab\cdot \mathbf{1}[1\mid am]+ab\cdot 1\cdot \mathbf{1}[ab\mid am]=ab+ab\cdot \mathbf{1}[b\mid m],$$

where we used $\gcd(a,b)=1$ to obtain $ab\mid am$ if and only if $b\mid m$. Similarly,

$$f_\mu(am)=a\cdot b\cdot \mathbf{1}[a\mid am]+b\cdot a\cdot \mathbf{1}[b\mid am]=ab+ab\cdot \mathbf{1}[b\mid m].$$

Hence $f_\lambda(am)=f_\mu(am)$ for all m. By the symmetric argument applied to bm, we get $g^\lambda_{a,b}=g^\mu_{a,b}$.

Case 2: $\gcd(a,b)=g\geq 2$. Take $\lambda=(g,1^g)$ and $\mu=(1^{2g})$ on $n=2g$ points. Since $g\mid a$ and $g\mid b$, we have $g\mid am$ for all m, so $f_\lambda(am)=g+g=2g=f_\mu(am)$. This is a constant collision. □

4. Parameter Classification

Theorem 4.1 (Parameter Classification)**.** *The transform $g_{a,b}$ is injective on cycle-closed multisets if and only if* $\min(a,b)=1$.

Proof. Sufficiency follows from Theorem 2.1; necessity from Theorem 3.1. The counterexamples in Theorem 3.1 are themselves cycle-closed, so no additional hypothesis is needed for necessity. □

5. The Rotation Obstruction

Theorem 5.1 (Rotation Obstruction)**.** *Let S be any multiset of complex numbers and $\omega\in\mathbb{C}$ with $\omega^{a+b}=1$. Then $g^{\omega S}_{a,b}=g^S_{a,b}$.*

Proof.

$$g^{\omega S}_{a,b}(m)=\sum_j(\omega\lambda_j)^{am}\cdot\sum_k(\omega\lambda_k)^{bm}=\omega^{(a+b)m}\cdot p_S(am)\cdot p_S(bm)=g^S_{a,b}(m).$$ □

The group μ_{a+b} acts on multisets by scalar multiplication, and $g_{a,b}$ is invariant under this action. If $\omega S\neq S$ for some $\omega\in\mu_{a+b}$, then S and ωS form a non-trivial collision pair.

Proposition 5.2. *Let S be cycle-closed and $\omega\in\mu_{a+b}$, $\omega\neq 1$. Then either $\omega S=S$ or ωS is not cycle-closed.*

Proof. A cycle-closed multiset $S=\bigsqcup_i\mu_{d_i}$ satisfies $\omega S=S$ if and only if $\mathrm{ord}(\omega)\mid\gcd(d_1,\dots,d_r)$. If $\omega S\neq S$, then at least one component μ_d maps to a coset $\omega\mu_d$ that is not a complete set of roots of unity for any order. Hence ωS is not cycle-closed. □

Corollary 5.3. *Within the class of cycle-closed multisets, the rotation obstruction produces only trivial collisions.*

6. The Galois Characterization

Definition 6.1. A multiset $S\subset\mu_N$ is *Galois-stable* if $\sigma_j(S)=S$ (as multisets) for every $\sigma_j\in\mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$, where $\sigma_j(\zeta_N)=\zeta_N^j$ for $\gcd(j,N)=1$.

Theorem 6.2 (Galois Characterization)**.** *A multiset $S\subset\mu_N$ is cycle-closed if and only if it is Galois-stable.*

Proof. The Galois orbit of ζ_N^i is the set Φ_d of primitive d-th roots of unity, where $d=N/\gcd(i,N)$. Hence Galois-stable multisets are precisely unions of complete sets Φ_d.

Cycle-closed multisets are unions of complete sets μ_d. Since $\mu_d=\bigsqcup_{e\mid d}\Phi_e$, every μ_d is a union of Φ_e's. Conversely, the sets $\{\Phi_e\}_{e\mid N}$ partition μ_N, and the sets $\{\mu_d\}_{d\mid N}$ are precisely the upward-closed unions of this partition under the divisibility order. Hence both collections generate the same Boolean algebra of subsets of μ_N, and the two conditions are equivalent. □

6.1. **The Galois obstruction.** For a non-Galois-stable multiset S, there exists σ_j with $\sigma_j(S) \neq S$. Since $g_{a,b}^{\sigma_j(S)}(m) = \sigma_j(g_{a,b}^S(m))$, the multisets S and $\sigma_j(S)$ produce distinct transforms whenever the values $g_{a,b}^S(m)$ do not all lie in the fixed field of σ_j. In particular, if $g_{a,b}^S(m) \notin \mathbb{Q}$ for some m, then some Galois automorphism moves it, producing a collision. More generally, a collision with $\sigma_j(S)$ is avoided only if $g^S(m) \in \mathbb{Q}(\zeta_N)^{\langle\sigma_j\rangle}$ for all m.

6.2. **The gap: why cycle-closed is not necessary.** For certain non-Galois-stable multisets, the product $p_S(m) \cdot p_S(km)$ can be rational for all m despite p_S itself being irrational-valued. In such cases, $\sigma_j(g^S) = g^S$ even though $\sigma_j(S) \neq S$, and no Galois-induced collision exists.

Example 6.3. In μ_7, let $S = \{\zeta_7, \zeta_7^6\} = \{\zeta_7, \bar{\zeta}_7\}$. This is not cycle-closed, since the only cycle-closed multisets in μ_7 are unions of $\{1\}$ and μ_7. However, $p_S(m) = 2\cos(2\pi m/7) \in \mathbb{R}$, and the transform $g_{1,2}^S(m) = 4\cos(2\pi m/7)\cos(4\pi m/7)$ takes values in $\mathbb{Q}(\cos(2\pi/7))$, a proper subfield of $\mathbb{Q}(\zeta_7)$.

Direct computation confirms that all six multisets $\{\zeta_7^j, \zeta_7^{7-j}\}$ for $j = 1, \ldots, 6$ produce pairwise distinct g-transforms. No collision occurs despite non-Galois-stability.

Remark 6.4. Computationally, for $N \leq 12$ and multiset sizes up to 5, every collision involves at least one multiset with non-rational g-values. No collision has been found among multisets with rational-valued g-transform. This suggests that the maximal domain of injectivity may be related to the rationality of the transform, but this remains unproved.

7. The Maximal Domain: An Open Problem

For $g_{1,k}$ acting on multisets in μ_N, three regimes are identified:

(a) *Cycle-closed multisets:* always distinguished (Theorem 2.1).
(b) *Non-Galois-stable with irrational g:* collision with $\sigma_j(S)$ for some Galois automorphism (Theorem 6.2 and Section 6.1).
(c) *Non-cycle-closed with rational g:* no collision observed for $N \leq 12$. These multisets escape both the rotation and Galois obstructions.

Problem 7.1. Characterize the maximal class $\mathcal{M}_{a,b}(N)$ of multisets in μ_N on which $g_{a,b}$ is injective.

From the evidence:

$$\text{cycle-closed} \subsetneq \mathcal{M}_{1,k}(N) \subseteq \{S : g_{1,k}^S(m) \in \mathbb{Q} \text{ for all } m\} \cup \text{cycle-closed}.$$

The gap between cycle-closed and the maximal domain reveals a subtle phenomenon: injectivity depends not on the symmetry of the multiset itself, but on the symmetry of the transform's values. A multiset can be asymmetric (non-Galois-stable) yet produce symmetric (rational) transform values through cancellation in the product $p_S(m) \cdot p_S(km)$. This is analogous to the distinction in invariant theory between an object being invariant and a function of that object being invariant.

8. Spectral Interpretation

8.1. **Synchronized tensor operators.** For a permutation matrix A, the operator $S_{a,b} = A^a \otimes A^b$ has spectrum determined by $g_{a,b}^S$ via Newton's identities.

Corollary 8.1. *The spectrum of $A^a \otimes A^b$ determines the cycle type of a permutation if and only if* $\min(a, b) = 1$.

8.2. **The Speed-1 Principle.** When $a = 1$, the transform $g_{1,k}(m) = f(m) \cdot f(km)$ contains f evaluated at the base frequency. The ratio $g(m)/g(km)$ telescopes because the shared factor $f(km)$ cancels. When $\min(a, b) \geq 2$, the function f is never probed below multiples of $\min(a, b)$, creating a permanent blind spot.

Remark 8.2. The descriptive complexity hierarchy FO $\subsetneq$ FO+TC$_1$ $\subsetneq$ FO+TC$_2$ = NL uses synchronized pointer traversal at speeds $(1, 2)$. The parameter classification provides spectral evidence for why one pointer must track the base dynamics directly. A formal connection remains to be established.

9. Computational Verification

All stated theorems have been verified computationally. The inversion formula (Theorem 2.1) recovers the cycle type exactly for all partitions of $n \leq 15$ and all speeds $k = 2, \ldots, 10$. The counterexample families of Theorem 3.1 have been confirmed for all tested parameter pairs. The Galois equivalence (Theorem 6.2) has been verified exhaustively for $N \leq 12$. The existence of non-cycle-closed multisets with no collision (Example 6.3 and Remark 6.4) has been confirmed for all $N \leq 12$ and multiset sizes up to 5.

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